Multiple Linear Regression: Interpretation, OVB, Heteroscedasticity, and Tests

Data Analytics II - Assignment 4

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- Keep your answers concise and precise.
- **Points are awarded only if the answer is both correct and within any stated sentence limit.** Answers that exceed the sentence limit will receive no points, even if the content is correct.
- There is no need to use R for this exercise since it tests your theoretical understanding. If helpful, you can use R for computations of course.

Part I [3.5 points]

The following output was obtained by estimating a model for household income on a cross-sectional sample of $N = 534$ households drawn from the 1991 Survey of Income and Program Participation (SIPP). Each household has at least one earner aged 25–64.

Variable	Description
`inc`	Annual household income (in USD).
`tw`	Total household wealth (in USD).
`fsize`	Household size (number of persons).
`age`	Age of the main earner (in years).
`educ`	Years of education of the main earner.
`marr`	Main earner is married (1 = yes, 0 = no).

The estimated model is:

$$\log(inc_i) = \beta_0 + \beta_1 \cdot fsize_i + \beta_2 \cdot \log(tw_i) + \beta_3 \cdot age_i + \beta_4 \cdot age_i^2 + \beta_5 \cdot educ_i + \beta_6 \cdot marr_i + u_i, \quad i \in \{1, \dots, N\}.$$

	Estimate	Std. Error
$\hat{\beta}_0$ (Intercept)	3.8914	0.4102
$\hat{\beta}_1$ (`fsize`)	-0.0412	0.0201
$\hat{\beta}_2$ (`log(tw)`)	0.2873	0.0198
$\hat{\beta}_3$ (`age`)	0.0512	0.0087
$\hat{\beta}_4$ (`age^2`)	-0.0006	0.0001
$\hat{\beta}_5$ (`educ`)	0.0724	0.0063
$\hat{\beta}_6$ (`marr`)	0.4217	0.0512

1. Coefficient interpretation [0.75 points]

Interpret the following coefficients carefully and quantitatively. Always include the *ceteris paribus* qualifier where appropriate.

(a) [0.25 points] Interpret $\hat{\beta}_1$ (`fsize`), $\hat{\beta}_2$ (`log(tw)`), and $\hat{\beta}_6$ (`marr`). For each of the three coefficients:

- State the interpretation in words,
- choose the economically correct interpretation formula from the following list and apply it correctly:
  i. $\hat{\beta}_j$
  ii. $\hat{\beta}_j \times 100$
  iii. $(e^{\hat{\beta}_j} - 1) \times 100$
- if more than one of these formulas could reasonably be considered, briefly justify your choice,
- if the approximate and exact percentage interpretations both apply and differ meaningfully, report both values,
- use at most two sentences per coefficient.

(b) [0.5 points] The model includes both `age` and `age^2`.

1. [0.25 points] Write down the marginal effect of `age` on $\log(inc)$. Evaluate this marginal effect at $age = 25$ and interpret it in one sentence.
2. [0.25 points] Compute the turning point age^* at which the marginal effect changes sign. Interpret the economic meaning of this turning point in one sentence.

2. Model specification [0.75 points]

A reviewer argues that `twoearn` (a dummy equal to 1 if two persons in the household are employed) was omitted from the model above. Assume for this question that the true model includes `twoearn` as an additional regressor with coefficient β_{tw}.

Use the omitted variable bias formula

$$\mathbb{E}[\breve{\beta}_j] = \beta_j + \underbrace{\beta_{tw} \cdot \delta_j}_{\text{bias}}, \qquad \delta_j \approx \frac{\text{Cov}(x_j, \, twoearn)}{\text{Var}(x_j)}.$$

Determine the **direction** of the bias on:

- $\breve{\beta}_1$ (`fsize`),
- $\breve{\beta}_6$ (`marr`).

For each coefficient, proceed in exactly three steps:

1. argue the sign of β_{tw},
2. argue the sign of δ_j,
3. conclude whether the estimated coefficient is biased upward or downward relative to the true coefficient.

Use at most one sentence per step and coefficient (that will give $2 \times 3 = 6$ sentences).

3. Residual patterns and inference [0.75 points]

Suppose you plot the OLS residuals $\hat{u}_i$ against `educ`. The cloud of points is narrow for households with less than 12 years of education, but much wider for households with 15 or more years of education.

(a) [0.25 points] Which Gauss–Markov assumption does this pattern call into question? State it formally in one sentence for this specific case of education.

(b) [0.25 points] If this assumption is violated but MLR.1–4 still hold:

1. Is OLS still unbiased?
2. Is OLS still BLUE?

Give a one-sentence justification for each answer.

(c) [0.25 points] What does this pattern imply for standard OLS standard errors and hypothesis tests if we still use the homoscedasticity-based formula? Answer in at most three sentences.

4. Inference for household size [0.75 points]

Conduct a **one-sided** test of

$$H_0 : \beta_1 \geq 0 \qquad \text{vs} \qquad H_1 : \beta_1 < 0$$

using a significance level of $\alpha = 1\%$.

Your answer must include:

1. the hypotheses,
2. the test statistic,
3. the observed t-value,
4. the reference distribution under H_0 (including degrees of freedom),
5. the critical value,
6. the p-value,
7. the test decision in words.

5. Testing age effects [0.5 points]

A student proposes testing the effect of age by looking separately at the t-tests for `age` and `age^2`.

1. Briefly explain why separate t-tests are not the correct way to test whether age has any effect jointly. (Max two sentences.)
2. Name the correct test for the joint null hypothesis

$$H_0 : \beta_3 = 0 \text{ and } \beta_4 = 0.$$

3. Write down the test-statistic in terms of SSR_r, SSR_{ur}, N, k, and q.

Part II [1.5 points]

6. Error Variance and OLS Precision [1.5 points]

You may use the following formulas:

Under homoscedasticity,

$$\mathrm{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

Under heteroscedasticity,

$$\mathrm{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.$$

(a) [0.5 points] In up to three sentences, explain the difference between these two formulas and what the matrix Ω represent in the heteroscedasticity formula.

(b) [0.25 points] Why can the homoscedastic OLS standard error formula be misleading when the residual variance is larger for highly educated households than for less educated households? Answer in at most three sentences and relate your answer to Question 3.

(c) [0.5 points] Under homoscedasticity, what does Ω simplify to? Write it down mathematically.

(d) [0.25 points] A fellow student says: "If OLS is unbiased, then the usual OLS standard errors are automatically valid." Is this statement correct? Answer yes or no and justify in at most two sentences.